LESSON 2.3

Describe the Data Using the Median

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the median of a data set and use it to describe what is typical.

WHY THIS MATTERS You're the stats analyst for the school basketball team. Coach needs a report on the team's scoring: in the last 7 games they scored 42, 38, 55, 41, 38, 44, and 40 points. Which average best represents the team's typical performance?

REMEMBER

Describe the Data Using the Median

Measures of Center: Median. Formula: Median = Middle value in an ordered set Mean, median, and mode are three ways to find the center, or "typical" value, of a data set. The mean is the average, the median is the middle, and the mode is the most common value.

1. Order all data values from least to greatest.
2. Odd count: The median is the exact center number.
3. Even count: The median is the mean of the two middle numbers

Words: **Median *(Mediana)*** **Mode *(Moda)*** **Outlier *(Valor atípico)*** **Data distribution *(Distribución de datos)*** **Variability *(Variabilidad)***

1 **Circle the letter of the best answer. Show how you know.**

A histogram bar for the interval 10–19 has a height of 6. How many data values fall in that interval?

(A) 6
(B) 10
(C) 16
(D) 19

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A histogram uses the intervals 0–9, 10–19 and 20–29. What is the width of each interval?

(A) 3
(B) 9
(C) 10
(D) 29

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A histogram has bar heights 3, 7, 5 and 2. How many data values are in the whole set?

(A) 4
(B) 7
(C) 15
(D) 17

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

How is a histogram different from a bar graph?

(A) Its bars show number intervals and touch each other
(B) Its bars are always taller
(C) It uses no numbers
(D) It shows only one category

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Your class took a math quiz. The scores were: 85, 92, 78, 88, 92, 95, 72, 88, 92, 84. The teacher says 'most students scored around 87.' A student disagrees and says 'the most common score was 92.' The teacher said 'most students scored around 87.' Which measure is that?

(A) The mode
(B) The mean (average)
(C) The median
(D) The range

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Your class took a math quiz. The scores were: 85, 92, 78, 88, 92, 95, 72, 88, 92, 84. The teacher says 'most students scored around 87.' A student disagrees and says 'the most common score was 92.' The student said 'the most common score was 92.' Which measure is that?

(A) The mean
(B) The median
(C) The mode
(D) The range

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The mean (43) is higher than the median (41). Why? Which better represents a typical game?

SENTENCE STARTER The mean is higher than the median because the high score of ___ pulls the mean ___. The ___ better represents a typical game because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A team scored 4, 4, 7, and 9 goals in four matches. What is the MODE?

(A) 4 (B) 6

(C) 7 (D) 9

Put the data in order _____

WORK

2 **Write the letter of the matching item on each line.**

Match each sports data set to the value of the measure named in CAPITALS.

_____ Game points 9, 3, 7, 5, 6 → MEAN

_____ Rebounds 8, 3, 11, 7, 1 → MEDIAN

_____ Jersey numbers 9, 2, 9, 6, 9, 3 → MODE

_____ Assists 14, 10, 18, 12, 16 → MEAN

(A) 6

(B) 7

(C) 9

(D) 14

PART 2 Apply it

real-world problems

3 Circle the letter of the best answer. Show how you know.

Your class took a math quiz. The scores were: 85, 92, 78, 88, 92, 95, 72, 88, 92, 84. The teacher says 'most students scored around 87.' A student disagrees and says 'the most common score was 92.' What is the median of the 10 scores?

(A) 86.6
(B) 87
(C) 88
(D) 92

Put the data in order _____

WORK

4 Circle the letter of the best answer. Show how you know.

A player scored 4, 8, 6, 10, and 2 points across five quarters. What is the MEAN number of points?

(A) 6
(B) 8
(C) 10
(D) 30

Put the data in order _____

WORK

5 Circle the letter of the best answer. Show how you know.

A goalie made 3, 7, 9, 2, and 5 saves in five games. What is the MEDIAN number of saves?

(A) 2
(B) 5
(C) 7
(D) 9

Put the data in order _____

WORK

6 **Circle the letter of the best answer. Show how you know.**

One game was a 55-point outlier. Which measure of center best shows a TYPICAL game?

(A) The mean, because it uses every score

(B) The mode, because it is always the typical value

(C) The range, because it shows the spread

(D) The median, because the outlier barely moves the middle value

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The mean (43) is higher than the median (41). Why? Which better represents a typical game?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help